NYC Taxi Analytics Dashboard

Business Intelligence Report

SQL Analysis using Google BigQuery & Dashboard Development in Looker Studio

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Prepared By

Ravsaheb Bansode

Data Analytics Project

Tools Used

- Google BigQuery
- SQL (Standard SQL)
- Looker Studio
- Google Cloud Platform (GCP)

Dataset:

NYC Yellow Taxi Trips 2015

Executive Summary

This project analyzes the NYC Yellow Taxi Trips 2015 dataset to understand taxi operations, customer travel behaviour, and revenue performance. The objective was to transform raw trip data into meaningful business insights that support informed decision-making.

The analysis began with SQL queries written in Google BigQuery. Each query was designed to answer a specific business question, such as identifying peak operating hours, analysing monthly revenue trends, comparing vendor performance, understanding payment preferences, and evaluating passenger travel patterns. Rather than extracting data alone, the focus was on solving real business problems through structured analysis.

The SQL results were then used to develop an interactive dashboard in Looker Studio. The dashboard presents key performance indicators including Total Trips, Total Revenue, Average Fare, Busiest Hours, Weekly Demand, Monthly Revenue Trend, Payment Method Distribution, Vendor Performance, Passenger Analysis, and Trip Distance Distribution.

The completed solution provides a clear view of business performance and demonstrates an end-to-end analytics workflow—from raw data exploration to business intelligence reporting.

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1. Project Overview

Taxi services generate a large volume of operational data every day. This data contains valuable information about customer demand, revenue, travel behaviour, and service performance. However, analysing millions of records manually is both time-consuming and impractical.

The purpose of this project was to analyse taxi trip data using SQL and convert the results into an interactive dashboard that helps stakeholders monitor business performance. The project follows a structured analytics approach, beginning with data exploration, followed by SQL-based analysis, and finally presenting the results through data visualization.

This project demonstrates practical skills in SQL, data analysis, dashboard development, and business intelligence by turning raw operational data into actionable insights.

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2. Business Objective

The project was developed to answer key business questions related to NYC taxi operations and provide meaningful insights for decision-makers.

The main objectives were:

- Analyse taxi trip data using SQL.
- Measure operational performance through key business metrics.
- Identify peak demand periods and travel trends.

- Evaluate monthly revenue performance.
- Understand customer payment preferences.
- Compare vendor performance.
- Analyse passenger behaviour and trip distance.
- Present analytical findings through an interactive dashboard.
- Support business decisions using data-driven insights.

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3. Project Workflow

The project followed a structured workflow consisting of five major stages.

Step 1 – Data Collection

The NYC Yellow Taxi Trips 2015 dataset was accessed directly from Google BigQuery Public Datasets.

Step 2 – Data Exploration

A sample of records was analysed to understand the dataset structure, identify important columns, and verify data quality before performing detailed analysis.

Step 3 – SQL Analysis

Business-focused SQL queries were written to answer questions related to demand, revenue, customer behaviour, payment methods, vendor performance, and trip patterns.

Step 4 – Dashboard Development

The SQL results were connected to Looker Studio, where interactive charts and KPI cards were created to present the findings in a simple and user-friendly format.

Step 5 – Business Insights

The completed dashboard was analysed to identify trends, evaluate business performance, and generate recommendations that could support operational improvements.

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4. Dataset Overview

4.1 Dataset Description

This project uses the **NYC Yellow Taxi Trips 2015** dataset available through **Google BigQuery Public Datasets**. The dataset contains detailed information about taxi trips completed across New York City during 2015.

Each record represents a single taxi trip and includes operational information such as pickup and drop-off time, trip distance, passenger count, fare amount, total amount, payment type, vendor details, and pickup location coordinates.

Since the dataset contains millions of records, it provides an excellent opportunity to perform large-scale data analysis and generate meaningful business insights using SQL.

4.2 Dataset Source

Dataset Name: NYC Yellow Taxi Trips 2015

Source: Google BigQuery Public Dataset

Platform: Google Cloud Platform (BigQuery)

The dataset is publicly available and can be queried directly using Standard SQL without downloading or manually importing the data.

4.3 Key Attributes Used

The following fields were used during the analysis:

Attribute	Description
pickup_datetime	Date and time when the trip started
dropoff_datetime	Date and time when the trip ended
passenger_count	Number of passengers
trip_distance	Distance travelled during the trip
fare_amount	Base fare charged
tip_amount	Tip paid by the customer
total_amount	Total amount paid for the trip
payment_type	Payment method used
vendor_id	Taxi vendor identifier
pickup_longitude	Pickup longitude
pickup_latitude	Pickup latitude

These attributes were selected because they directly support the business questions addressed in this project.

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5. Tools and Technologies

The project was completed using the following tools.

Tool	Purpose
Google BigQuery	Data storage and SQL query execution
SQL (Standard SQL)	Data exploration and business analysis
Looker Studio	Interactive dashboard creation
Google Cloud Platform	Platform for accessing BigQuery datasets

Each tool played a specific role in the analytics workflow, from querying raw data to presenting insights through visualizations.

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6. Methodology

The project followed a structured analytics process to ensure that every dashboard visualization was supported by meaningful analysis.

Step 1 – Data Exploration

The analysis began by examining a sample of records from the dataset. This helped in understanding the available columns, verifying data formats, and identifying the information required for further analysis.

Step 2 – Business Question Identification

Instead of creating visualizations directly, the key business questions were identified first. These questions focused on operational performance, customer behaviour, revenue trends, payment methods, vendor comparison, and travel patterns.

Step 3 – SQL Analysis

SQL queries were written to answer each business question individually. Aggregate functions, date functions, conditional logic, and filtering techniques were used to calculate business metrics and identify trends.

Step 4 – Dashboard Development

After validating the SQL results, the outputs were connected to Looker Studio to create an interactive dashboard. Appropriate visualizations were selected based on the type of analysis, making the dashboard easy to understand for both technical and non-technical users.

Step 5 – Business Interpretation

The final step involved interpreting the dashboard results to identify business trends and prepare recommendations that could help improve operational efficiency and decision-making.

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7. Why SQL Before Dashboard?

One of the key objectives of this project was to perform **analysis before visualization**.

Rather than creating charts directly from the raw dataset, SQL was first used to answer specific business questions. This approach ensured that each dashboard component was built on validated analytical results rather than simple data summaries.

Using SQL before dashboard development provided several advantages:

- Improved understanding of the dataset.
- Validation of business metrics before visualization.
- More accurate and meaningful KPIs.
- Faster dashboard development with pre-processed results.
- Better alignment between business questions and dashboard visuals.

This workflow reflects the approach commonly followed by data analysts in real-world business intelligence projects, where analysis forms the foundation for effective reporting and decision-making.

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8. SQL Analysis

The dashboard was built only after completing the SQL analysis. Each query was written to answer a specific business question and generate insights that could support business decisions. This approach ensured that every visualization displayed in the dashboard was based on validated analytical results rather than raw data.

Query 0 – Sample Data Preview

Business Question

What information is available in the dataset, and is it suitable for analysis?

Approach

A sample of ten records was retrieved from the dataset to understand the available columns, data types, and overall data structure before performing detailed analysis.

Key Observation

The dataset contains all the required information for analysis, including pickup and drop-off times, trip distance, fare amount, and total payment. This confirmed that the dataset was suitable for answering the business questions defined for the project.

Business Value

Exploring the dataset before analysis helped in selecting the correct attributes and reduced the possibility of errors during SQL development.

Query 1 – Peak Hours Analysis

Business Question

Which hours of the day experience the highest taxi demand?

Approach

The pickup timestamp was grouped by hour to calculate the total number of trips and the average revenue generated during each hour. The results were then ranked based on trip volume.

Key Observation

The analysis identified specific hours during which taxi demand was significantly higher than the rest of the day. These periods represent peak operating hours for the taxi service.

Business Value

Understanding peak demand helps management:

- Schedule drivers more efficiently.
- Reduce passenger waiting time.
- Improve fleet utilization.
- Support demand-based pricing strategies.

The results of this analysis were visualized in the **Busiest Hours of the Day** chart on the dashboard.

Query 2 – Weekly Trend Analysis

Business Question

How does taxi demand vary across different days of the week?

Approach

Trips were grouped according to the day of the week using the pickup date. Along with trip count, the average trip distance for each day was also calculated.

Key Observation

The results show noticeable variation in trip volume throughout the week, indicating that customer demand is influenced by weekday and weekend travel patterns.

Business Value

This analysis enables the business to:

- Plan weekly driver schedules.
- Allocate resources based on expected demand.
- Identify lower-demand days for promotional campaigns.

The findings are presented in the **Trips by Day of Week** visualization within the dashboard.

Query 3 – Monthly Revenue Analysis

Business Question

Which months contribute the highest revenue?

Approach

Trips were grouped by month to calculate the total number of trips, total revenue, and average fare for each month. The months were then ranked based on revenue generated.

Key Observation

The analysis shows that revenue remains relatively consistent throughout the year, with some months performing slightly better than others. This indicates stable customer demand with moderate seasonal variation.

Business Value

Monthly revenue analysis helps management to:

- Monitor financial performance.
- Identify seasonal business trends.
- Support budgeting and revenue forecasting.
- Plan marketing activities during lower-performing periods.

The dashboard presents these insights through the **Monthly Revenue Trend** chart.

Query 4 – Payment Type Analysis

Business Question

Which payment methods are most commonly used by customers?

Approach

Payment type codes were converted into meaningful payment method names using a CASE statement. The query then calculated the total number of trips and average tip amount for each payment method.

Key Observation

The analysis shows that Credit Card and Cash account for the majority of taxi payments, while other payment methods are used much less frequently.

Business Value

Understanding payment preferences helps the business to:

- Improve payment infrastructure.
- Encourage digital payment adoption.
- Analyse customer payment behaviour.
- Evaluate tipping patterns across payment methods.

These findings are displayed in the **Payment Method Split** section of the dashboard.

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9. SQL Analysis (Part 2)

The remaining SQL queries focus on operational performance, customer travel behaviour, vendor comparison, and location-based analysis. Together, these analyses provide a more comprehensive understanding of taxi operations and support data-driven business decisions.

Query 5 – Trip Distance Analysis

Business Question

What is the typical distance travelled by customers?

Approach

The query calculated the minimum, maximum, and average trip distance after filtering out invalid records with zero or unrealistic distances. This ensured that the analysis reflected actual customer travel patterns.

Key Observation

Most taxi trips cover relatively short distances, while long-distance trips are less frequent. This indicates that the taxi service is primarily used for short urban journeys.

Business Value

Understanding trip distance helps businesses to:

- Estimate operational costs.
- Improve fleet planning.
- Analyse customer travel behaviour.
- Support pricing strategy decisions.

The findings are presented in the **Trip Distance Distribution** chart on the dashboard.

Query 6 – Vendor Performance Comparison

Business Question

How do different taxi vendors perform in terms of trips and revenue?

Approach

The data was grouped by Vendor ID to calculate the total number of trips, total revenue, average trip distance, and average fare for each vendor. The results were ranked based on revenue.

Key Observation

The analysis shows a clear difference in revenue generated by the two vendors, indicating variations in operational performance and market contribution.

Business Value

Vendor comparison helps management to:

- Monitor vendor performance.
- Identify high-performing operators.
- Benchmark operational efficiency.
- Support performance-based business decisions.

The dashboard displays this comparison using the **Vendor Performance by Revenue** chart.

Query 7 – Top Pickup Locations

Business Question

Which locations generate the highest number of taxi pickups?

Approach

A spatial join was performed between the taxi trip dataset and the taxi zone dataset using geographic coordinates. Only valid location records were considered, and the analysis focused on trips recorded during January.

Key Observation

The analysis identified the most active pickup zones, highlighting areas where customer demand is consistently high.

Business Value

This information can help businesses to:

- Position drivers in high-demand locations.
- Reduce passenger waiting time.
- Improve fleet availability.
- Support location-based operational planning.

Note: This analysis was performed during the SQL phase but was not included in the final dashboard to maintain a clean layout and focus on the most impactful business metrics.

Query 8 – Average Fare by Passenger Count

Business Question

Does the number of passengers affect the average fare?

Approach

Trips were grouped according to passenger count, and the average fare and average trip distance were calculated for each group. Only valid passenger counts were included in the analysis.

Key Observation

The results show only minor differences in average fare across passenger groups, suggesting that fare values are influenced more by trip characteristics than by the number of passengers.

Business Value

Passenger analysis helps businesses to:

- Understand customer travel behaviour.
- Evaluate pricing consistency.
- Explore opportunities for shared ride services.
- Monitor changes in passenger trends over time.

The results are visualized in the **Average Fare by Passenger Count** chart.

SQL Analysis Summary

A total of **nine SQL queries** were executed during this project, each designed to answer a specific business question. The analysis covered demand patterns, revenue performance, payment methods, trip distance, passenger behaviour, vendor comparison, and location analysis.

Rather than building the dashboard directly from raw data, SQL was first used to validate business metrics and generate meaningful insights. This approach ensured that every visualization presented in the dashboard was supported by accurate analytical results and aligned with the project's business objectives.

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10. Dashboard Analysis

After completing the SQL analysis, the validated results were visualized in **Looker Studio** to create an interactive dashboard. The dashboard provides a consolidated view of operational and financial performance, enabling users to monitor key metrics and identify trends without analysing raw data. Each visualization was designed to answer a specific business question and support informed decision-making.

10.1 Key Performance Indicators (KPIs)

Three KPI cards are displayed at the top of the dashboard to provide an immediate overview of business performance.

KPI	Value	Business Significance
Total Trips	100,000	Indicates the total number of completed taxi trips analysed.
Total Revenue	₹589,655.82	Represents the total revenue generated from all trips.
Average Fare	₹4.29	Shows the average fare earned per trip.

These KPIs provide a quick summary of operational activity and financial performance before exploring the detailed visualizations.

10.2 Busiest Hours of the Day

This bar chart displays taxi demand across different hours of the day. It helps identify the periods with the highest trip volume, allowing operators to understand when demand is at its peak.

Business Insight

- Peak hours indicate when additional drivers may be required.
- Better shift planning can improve service availability.

- Fleet utilization can be optimized by aligning driver schedules with customer demand.

10.3 Trips by Day of Week

This visualization compares the number of trips completed on each day of the week. It highlights weekly demand patterns and shows how customer travel behaviour changes throughout the week.

Business Insight

- Supports weekly workforce planning.
- Identifies high-demand and low-demand days.
- Helps optimize driver allocation across the week.

10.4 Monthly Revenue Trend

The line chart illustrates revenue generated across different months of the year. It provides a clear view of revenue trends and helps identify seasonal variations.

Business Insight

- Enables management to monitor financial performance over time.
- Supports revenue forecasting and budgeting.
- Helps identify months that may require promotional campaigns or operational adjustments.

10.5 Payment Method Split

This chart shows the distribution of payment methods used by customers. It provides an overview of customer payment preferences and the relative share of each payment type.

Business Insight

- Demonstrates strong adoption of Credit Card and Cash payments.
- Supports decisions related to payment infrastructure.
- Helps evaluate opportunities to promote digital payment methods.

10.6 Average Fare by Passenger Count

This chart compares the average fare across different passenger groups. It helps determine whether the number of passengers has a significant impact on fare values.

Business Insight

- Average fare remains relatively consistent across passenger counts.
- Indicates that fare values are driven more by trip characteristics than by the number of passengers.
- Supports pricing analysis and customer behaviour studies.

10.7 Vendor Performance by Revenue

This visualization compares total revenue generated by each taxi vendor. It allows stakeholders to evaluate vendor contribution and identify performance differences.

Business Insight

- Simplifies vendor performance comparison.
- Helps identify high-performing vendors.
- Supports operational benchmarking and performance monitoring.

10.8 Trip Distance Distribution

The distribution chart illustrates how trip distances are spread across the dataset. It provides insight into the most common travel distances completed by customers.

Business Insight

- Indicates that most taxi trips are short-distance journeys.
- Helps businesses understand customer travel behaviour.
- Supports fleet deployment and pricing strategy decisions.

Dashboard Summary

The dashboard brings together multiple operational and financial metrics into a single reporting interface. Instead of reviewing separate reports, users can quickly evaluate business performance through interactive visualizations and KPI cards.

The combination of SQL analysis and dashboard development transforms raw trip data into meaningful business information. This enables stakeholders to monitor performance, identify trends, and make informed decisions based on reliable data.

Overall, the dashboard serves as an effective business intelligence solution by presenting complex analytical results in a clear, simple, and user-friendly format.

11. Key Findings

The SQL analysis and dashboard visualizations provided several insights into taxi operations, customer behaviour, and revenue performance. The most significant findings are summarized below.

11.1 High Trip Volume

The dashboard records **100,000 completed trips**, indicating strong demand for taxi services during the analysed period. This metric reflects the overall operational scale of the business and serves as the foundation for further performance analysis.

11.2 Consistent Revenue Performance

The taxi service generated a total revenue of **₹589,655.82**, with an average fare of **₹4.29** per trip. Monthly revenue remained relatively stable throughout the year, suggesting consistent customer demand with only minor seasonal fluctuations.

11.3 Peak Demand During Specific Hours

Trip demand is concentrated during particular hours of the day, indicating clear peak operating periods. Understanding these demand patterns can help improve fleet utilization and driver scheduling.

11.4 Variation in Weekly Demand

The number of trips differs across the days of the week, showing that customer travel behaviour changes depending on the day. This insight can support more effective workforce planning.

11.5 Customer Payment Preferences

Most customers prefer paying by **Credit Card** or **Cash**, while other payment methods account for only a small percentage of transactions. This highlights the importance of maintaining efficient digital and cash payment systems.

11.6 Difference in Vendor Performance

The vendor comparison shows that one vendor generated higher revenue than the other. Regular monitoring of vendor performance can help identify operational strengths and improvement opportunities.

11.7 Passenger Count Has Limited Impact on Fare

The analysis shows only minor differences in average fare across passenger groups, indicating that fare values are influenced primarily by trip characteristics rather than the number of passengers.

11.8 Short-Distance Trips Dominate

Most taxi journeys fall within shorter distance ranges, indicating that the service is primarily used for local travel rather than long-distance transportation.

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12. Business Recommendations

Based on the findings of this analysis, the following recommendations can help improve operational efficiency and business performance.

Optimize Driver Scheduling

Driver availability should be aligned with peak demand hours to reduce passenger waiting time and improve service levels.

Improve Weekly Resource Planning

Since trip demand varies across the week, staffing levels should be adjusted according to expected demand to maximize efficiency.

Strengthen Digital Payment Experience

As digital payments account for a significant share of transactions, improving payment reliability and offering customer incentives can further increase adoption.

Monitor Vendor Performance

Regular evaluation of vendor performance can help identify best practices, improve operational efficiency, and ensure consistent service quality.

Use Revenue Trends for Planning

Monthly revenue trends should be reviewed regularly to support budgeting, demand forecasting, and marketing initiatives during lower-performing periods.

Continue Dashboard Monitoring

The dashboard should be updated regularly so that management can monitor key performance indicators and identify changes in business performance at an early stage.

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13. Conclusion

This project demonstrates a complete business intelligence workflow, starting with raw data analysis and ending with an interactive dashboard.

The analysis began with data exploration in Google BigQuery, followed by SQL queries that answered important business questions related to demand, revenue, payment methods, vendor performance, passenger behaviour, and trip distance. These analytical results formed the foundation for developing the dashboard in Looker Studio.

The dashboard combines key performance indicators with interactive visualizations, enabling users to monitor operational and financial performance from a single interface. Rather than presenting raw data, it transforms complex information into clear and actionable insights that support business decision-making.

Overall, this project demonstrates practical skills in SQL, business analysis, data visualization, and dashboard development. It reflects an end-to-end analytics approach that can be applied to real-world business scenarios and provides a strong foundation for data-driven decision-making.

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14. Future Enhancements

The current dashboard provides a comprehensive overview of taxi operations; however, several enhancements could further improve its analytical capabilities.

- Add interactive filters for month, vendor, and payment method.
- Include pickup and drop-off location maps for geographic analysis.
- Develop real-time dashboards using live data sources.
- Build predictive models to forecast trip demand and revenue.
- Expand the analysis to compare multiple years of taxi data.
- Introduce automated dashboard refresh schedules for continuous reporting.

These improvements would increase the dashboard's analytical value and provide deeper insights for business users.

References

1. Google BigQuery Public Dataset – NYC Yellow Taxi Trips 2015.
2. Google BigQuery Documentation.
3. Google Cloud Platform Documentation.
4. Looker Studio Documentation.
5. NYC Taxi and Limousine Commission (TLC).